

Plant Pest Diseases Detection System in Rice Field Based on Aerial Mapping using Deep Learning

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Abstract— The global agricultural sector plays a crucial role in sustaining food security amidst population growth. However, traditional farming methods persist, highlighting the need for innovative approaches to improve crop mapping efficiency, especially in expansive rice fields. Indonesia, predominantly agrarian, faces challenges in becoming a top rice exporter due to high domestic consumption. Monitoring rice crop growth across extensive agricultural areas burdens farmers managing multiple fields. This study proposes a novel method utilizing cost-effective UAV technology for wide-area rice field imaging, using RGB imagery to derive NDVI without relying on specialized Near-Infrared (NIR) sensors. Traditional NDVI calculations typically require expensive multispectral or hyperspectral cameras, posing financial constraints and technical expertise demands. Advanced UAV technology facilitates efficient crop monitoring, height estimation, and soil analysis, integrated with diverse data collection tools and deep learning-based image processing, offering practical solutions for farmers aiming to enhance productivity and sustainability. The results of the deep learning YOLOv5m model produced an accuracy rate of 0.988 which detected rice fields planted with rice plants.

Keywords— *Deep Learning, Plant Disease Pest, Agriculture, Drone, Aerial Mapping*

I. INTRODUCTION

The agricultural sector plays an important role in providing food for the growing global population. Currently, agricultural machinery technology still uses many traditional approaches, so the development of innovative rice crop mapping systems is urgently needed to increase productivity in large agricultural areas. In addition, rice is a very important agricultural product, especially in Asian countries such as Japan, Thailand, and Indonesia, where it is a daily staple food for many people [1]. Indonesia, a country where most of the population are farmers, has a labour force of 88.89 percent by 2022 [2], which contributes to the country's agricultural production. Regular monitoring of the growth and development of rice plants in a large area is a heavy burden for farmers, who often have more than one farm.

This research proposes a novel approach to obtain images of paddy field area using cost-effective UAVs, converting RGB images into NDVI representations without relying on specialized Near Infrared (NIR) sensors. Traditionally, NDVI calculation uses expensive multispectral or hyperspectral cameras, which is a major challenge due to the high cost and technical expertise required. UAVs have advanced rapidly in the context of agriculture, being used for crop monitoring, crop height estimation, and soil and land analysis. The integration of drones with various data collection tools and methods, as well as image processing using deep learning, can

provide convenience for farmers. The use of RGB vegetation indices from UAV imagery offers a cost-effective and efficient solution for monitoring rice crops, becoming a viable alternative when multispectral imagery is not available [3]. Current crop health monitoring techniques utilize UAV and RGB imagery with synthetic NDVI methods such as ExG, ExR, ExGr, NGRDI, IKAW, MGRVI, and GLI [3].

This monitoring system allows evaluation of the greenness of rice leaves, which reflects the chlorophyll content in the plant. With cost-effective UAVs, farmers can monitor rice plant growth more efficiently and accurately through RGB images converted to NDVI without specialized Near Infrared (NIR) sensors.

We collected data from farms in Sidoarjo, Gresik and Yogyakarta, Indonesia, to support this research. With this innovative approach, it is expected to open up new opportunities in rice crop growth monitoring in Indonesia, improving efficiency, reducing costs, and producing more accurate results. This not only supports agricultural sustainability, but also the potential for increased productivity for a country with a significant farmer population like Indonesia.

II. RELATED WORKS

Joseph K. Mhango, et al. [4] conducted research to detect plants and estimate the density of plants built using models from the Convolutional Neural Network based on the Faster Region (FRCNN), as well as images taken from aerial images of drones made two-dimensional maps accurately.

Estrada-Peraza, et al. [5] conducted research to identify coffee leaf rust disease (CLR, *Hemileia vastatrix*), using tools such as vegetation index (VI). An algorithm was designed to identify CLR based on VI and achieved 94.5% effectiveness in detecting CLR. Ke Li1, et al. [6] conducted research for disease identification in rice using deep learning with the improved YoloV5s method, which reduced the backbone network calculation, reduced the model weight file to 1/4 of the original model, and accelerated the prediction speed. The model can distinguish and successfully detect rice leaf diseases. Anjanadevi B, et al [7] conducted research to detect healthy and sick tomato leaves using deep learning by comparing several architectures of Mobile Net, Dark Net-19, and ResNet-101. Muhammad Hammad Saleem, et al. [8] conducted research into the visualizing various plant diseases by identifying in detecting diseases, even symptoms are not clearly visible using deep learning methods.

In this research, we propose to build on the work of [4]-[8] and focus on plant disease detection using YOLOv5 and drones.

TABLE I. COMPARISON WITH RESEARCH PREVIOUS

Previous Studies	Object	Method	Drone
[4]	Destiny of plants	FRCN and Aerial Mapping	✓
[5]	Disease detection in coffe	Image and vegetation index	✗
[6]	Disease detection in paddy	YOLOv5 + Improve Backbone	✗
[7]	Disease detection in Tomato	Mobile Net, Dark Net-19, and ResNet-01	✗
[8]	Disease detection	Image and Deep Learning	✗
Proposed System	Disease detection	Yolov5, Vegetation Index, and Aerial Mapping	✓

III. SYSTEM DESIGN

This section provides an overall view of the research framework, which includes several key components: (1) Design System; (2) Data Acquisition; (3) Mapping; (4) modeling using YOLOv5; (5) Non-neural synthetic NDVI approach; (6) testing with image similarity metrics.

A. Design System

Proposed system for plant pest and disease detection using drones and the YOLOv5 method is illustrated in Fig. 1.

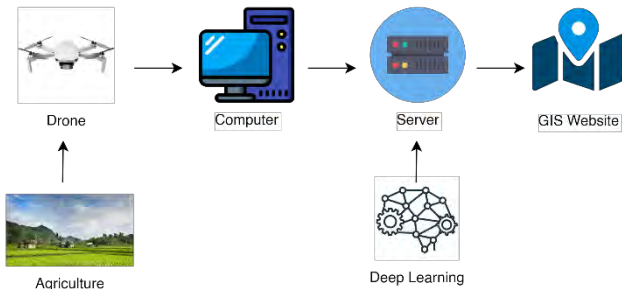


Fig. 1. Design System

The system in Fig. 1 show device used to collect data is a drone. Where the drone will be flown on agricultural land and the drone camera will take images of rice leaves. The data is processed by the Yolo method to classify diseases detected on rice leaves and will be uploaded to the server via the internet. The successfully detected data will be stored in the database and displayed on the webGIS where it serves as an easy-to-use interface to visualize the detected data on the map.

In this research, as seen in the flowchart in Fig. 2. It shows process begins with the drone operating over the rice field area to be mapped. The drone performs mapping by collecting geographic data and imagery of the mapped area. Once the mapping is complete, the drone takes pictures of the area. The captured images are then transferred to a laptop for further processing. At the next stage, the transferred images are processed using the YOLOv5 model to perform object or area detection in the images. The result of this detection is then displayed in the form of a segmented image, which shows the detected object or area. Once all the steps are completed, the process is terminated.

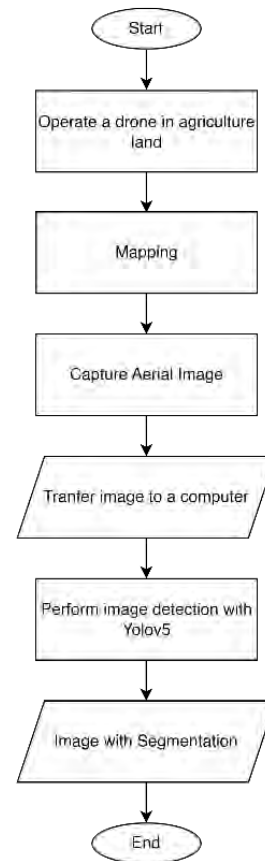


Fig. 2. Flowchart System

B. Data Acquisition

In this study, image data was collected using two types of UAVs: DJI Phantom 4 and DJI Mavic Air 2S, both of which are equipped with RGB cameras. In addition, we used a Mapir Survey 3 camera with an RGN channel to capture NDVI images. These two types of RGB UAVs were used to enrich the variety of image characteristics as they have different RGB camera specifications. Data were collected from rice fields in Gresik and Yogyakarta, Indonesia, with the UAVs flown at an altitude of 25 to 30 meters. The data collected was in the form of photographs. In this study, we also compared the synthetic NDVI from several approaches with the real NDVI using test data. After the capture image is performed.

C. Mapping

Mapping with drones begins with planning the flight path to ensure the entire area is covered with sufficient image overlap. Drones equipped with high-resolution cameras are flown automatically by flight planning software, taking successive images along a predetermined path. These images are then processed with software such as Agisoft Metashape to produce an orthomosaic, an accurate 2D map that combines all the images with geometry corrections. This orthomosaic is used for farmland segmentation, where land is identified and classified using image processing and machine learning techniques, aiding in efficient farmland analysis and management.

D. YOLOv5

YOLOv5 (You Only Look Once version 5) is a deep learning model used for object detection. In addition to object detection, YOLOv5 also supports instant segmentation, which is a technique to identify and determine the shape of objects

in an image more precisely, generating a mask for each detected object.

There are four versions of the YOLOv5: YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. Among them, the model size of YOLOv5s is about one-tenth of that of YOLOv4, with faster recognition and location speed, while the accuracy is not inferior to YOLOv4. Since this research requires real-time performance and lightweight models, we plan to improve the network architecture of YOLOv5s based on thorough consideration of accuracy, efficiency, and model size [6].

E. Non-neural synthetic NDVI approach

In our study, we also replicated an existing strategy that utilizes manipulation between image channels to produce synthetic NDVI depictions. We have combined various techniques from previous studies, with the formula outlined in Table 2 This method enables the extraction of synthetic NDVI from RGB images without the need for neural networks, thus offering an alternative for environmental and agricultural screening. We have re-applied this methodology using our dataset.

TABLE II. FROMULA SYNTHETIC NDVI USING RGB IMAGE

Method	Formula
Exg	$ExG = 2 * Green - Red - Blue$
Exgr	$ExGr = Exg - 1.4 * Red - Green$
GLI	$GLI = \frac{2 * Green - Red - Blue}{2 * Green + Red + Blue}$

F. Image similarity assessment

Assessing the similarity between the generated image and the actual NDVI image is crucial for performance evaluation. We use various metrics to thoroughly examine the quality of the generated image. Normalized Root Mean Square Error (NRMSE), which is calculated using (1), measures the average magnitude of pixel differences between two images.

$$NRMSE = \frac{\sqrt{\sum_{i=1}^n (y_i - \hat{y})^2}}{y_{max} - y_{min}} \quad (1)$$

Furthermore, we use the Peak Signal-to-Noise Ratio (PSNR) to assess the image quality and reconstruction accuracy. PSNR, which is often used in image compression, measures the ratio between the highest signal strength and the intensity of noise interference, which is calculated by (3) [9]. To measure PSNR, we must first calculate the Mean Squared Error (MSE) using (2). MSE measures the mean squared difference between the estimated pixels and the actual pixels [9].

$$MSE = \frac{1}{HW} \sum_{p=0}^H \sum_{q=1}^W [y(p, q) - \hat{y}(p, q)]^2 \quad (2)$$

$$PSNR = 10 \log_{10}(\max Val^2) / MSE \quad (3)$$

In addition to pixel-based metrics, we use the Oriented FAST and Rotated BRIEF (ORB) feature-based metrics. This method identifies and matches key points between the generated image and the reference image by using the ORB algorithm. The similarity score is calculated as the ratio of matched areas to the total number of matches.

In addition, we used the Structural Similarity Index (SSIM), a metric designed to reflect human visual perception.

In contrast to pixel-based metrics, SSIM considers changes in structural information, perceived brightness, and contrast, to measure the structural similarity between two images. This assessment is done by taking luminance, contrast, and structure into account, and is calculated using (4).

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_x\sigma_y + c_2)}{(\mu_x^2\mu_y^2 + c_1)(\sigma_x^2\sigma_y^2 + c_2)} \quad (4)$$

IV. EXPERIMENTAL RESULT

A. Datasets

The data obtained from aerial mapping using unmanned aircraft taken from various rice fields such as rice fields in Gresik, Sidoarjo, and Yogyakarta are used for the needs of the training model process and improve the accuracy of prediction model as in Fig. 3 data obtained as many as 805 images.



Fig. 3. Datasets

B. Labeling and Augmentation

Before training annotations are performed to identify regions and add important information aimed at identifying objects, or areas as classes, these annotations are important to provide insight into the model. RoboFlow is a tool to annotate the dataset as shown in Fig. 4.

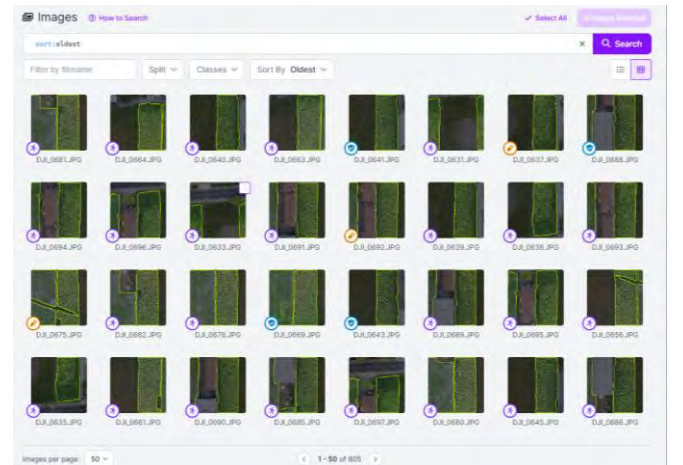


Fig. 4. Annotate the datasets using RoboFlow

Larger datasets make it easier for the model to recognize and detect objects in various situations, so proper labeling is crucial in developing an effective model. To improve the performance of the YOLOv8 model, a large enough dataset is

required to help reduce the risk of overfitting and improve the accuracy of the results displayed by the model. However, since the number of available datasets is still limited, efforts to enlarge the dataset are essential to improve the generalization and performance of the object detection model.

C. Split the dataset; Train, Valid, Test

The division of the dataset into training, validation, and test data, as shown in Fig. 5, is done with a proportion of 70% for training data, 20% for validation data, and 10% for test data. This division is done because most of the training data is used to train the model and optimize its parameters, the model can learn the patterns in the data. Validation data is used to evaluate the performance of the model during training. With periodic validation, measure the model's ability to generalize and detect overfitting or underfitting. Test data, on the other hand, is used to assess the final performance of the model after training is complete. This test data has never been seen by the model before and is used to provide an objective estimation of the model's performance in the real world.



Fig. 5. Split the datasets

D. Training Model: YOLOv5

Before training the dataset, the first step is to determine the most suitable YOLOv5 base model. Although the architecture of the five versions of the YOLOv5 base model is identical, the difference lies in the number of channels or filters in each layer as well as the depth of the model [10]. YOLOv5 includes five different versions: YOLOv5n, YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x. Table 3 shows a comparison of the performance and size of each YOLOv5 version on a 640 x 640 pixel image [10]. Larger YOLOv5 models offer higher accuracy, but the training process takes longer. YOLOv5x is the largest model with the highest accuracy, while YOLOv5n is the simplest with the lowest FLOP. The choice of model depends on the need for accuracy, speed, and available resources.

TABLE III. YOLOV5 INSTANCE SEGMENTATION MODELS

Models	mAP (box)	mAP (mask)	Inference Speed (ms)	Parameters (M)
YOLOv5n-Seg	27.6	23.4	1.2	2.0
YOLOv5s-Seg	37.0	31.7	1.4	7.6
YOLOv5m-Seg	45.0	37.1	2.2	22.0
YOLOv5l-Seg	49.0	39.9	2.9	47.9
YOLOv5x-Seg	50.7	41.4	4.5	88.8

hyperparameters as in the test to train this basic model: image size of 640 pixels, epoch of 50, and batch of 16. The test results indicated that YOLOv5m achieved the highest mAP value of 98 percent and the highest accuracy of 96 percent, as shown in Fig. 6, Fig. 7, and Fig. 8. Compared to the other configuration models, YOLOv5s is the most basic model with the least number of parameters, while YOLOv5x is the most complex configuration model.

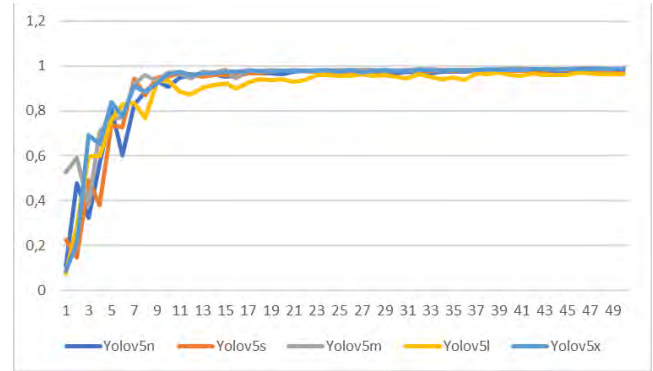


Fig. 6. Comparison of mAP50 values from the YOLOv5

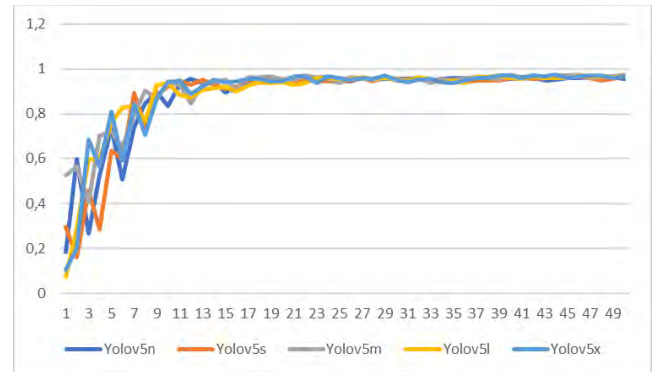


Fig. 7. Comparison of precision values from the YOLOv5

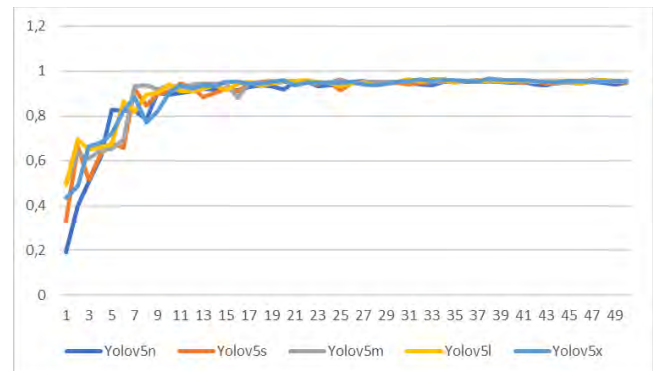


Fig. 8. Comparison of recall values from the YOLOv5

Three models, it can be seen that they have almost similar metric values, but YOLOv5m has the highest matrix evaluation value. This could be influenced by various parameters such as model design, number of datasets, hyperparameters, training method, and test data used. Therefore, this study recommends the use of YOLOv5m as it is suitable for detecting agricultural land, especially those planted with rice crops and requires fast detection.

The test is carried out after obtaining the right yolov5 model configuration model, namely yolov5m, after obtaining the right model, a retest is carried out with 50 epochs. This decision is based on the observation that the number of epochs directly affects the accuracy of the model after repeated testing. Fig. 9 shows the decreasing loss values for bounding box prediction, object segmentation, and object detection (train/box_loss, train/seg_loss, and train/obj_loss).

This indicates that the model gets better at predicting accurate bounding boxes, segmenting objects, and detecting the presence of objects as the number of epochs increases.

However, the loss value for object classification (train/cls_loss) remains 0, indicating that no object classification is performed or only one class is detected. The precision and recall graphs for bounding box and mask segmentation (metrics/precision(B), metrics/recall(B), metrics/precision(M), and metrics/recall(M)) show good performance with high precision and recall values, indicating that the model rarely produces false positives and false negatives.

Validation graphs at Fig. 9 (val/box_loss, val/seg_loss, val/obj_loss, and val/cls_loss) also show a decrease in the loss value, which reflects the ability of the model to generalize well to data not seen during training. In addition, the high mean average precision (mAP) values at various IoU thresholds (metrics/mAP_0.5(B), metrics/mAP_0.5:0.95(B), metrics/mAP_0.5(M), and metrics/mAP_0.5:0.95(M)) indicate that the model is able to detect and segment well at various levels of precision. Overall, these graphs show that the YOLOv5 model has significantly improved its performance in object detection and segmentation during both training and validation, which is evident from the decreased loss values as well as the high precision, recall, and mAP values.

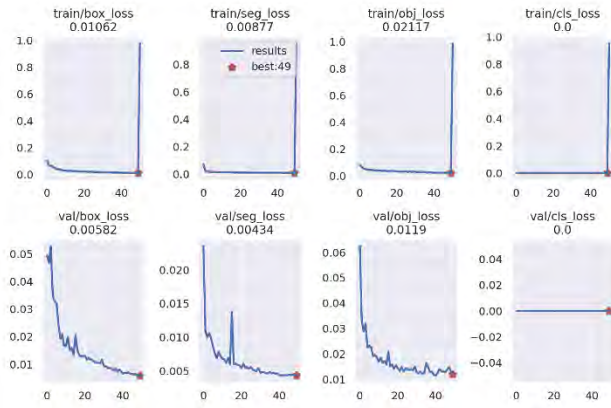


Fig. 9. Evaluation train for YOLOv5m training

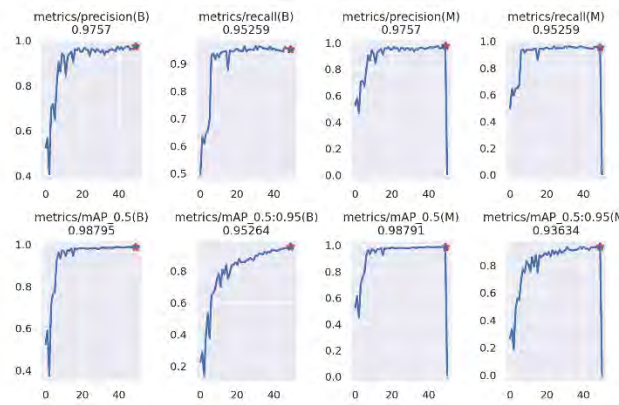


Fig. 10. Evaluation matrix for YOLOv5m training

E. Comparison with the synthetic NDVI non-neural approach

We evaluated existing methods and our proposed method using our dataset, and then compared it with the actual NDVI value.

TABLE IV. AVERAGE SCORE OF PERFORMANCE METRICS ON TEST DATA TOWARDS TRUE NDVI

Method	NRMSE	PSNR	ORB	SSIM
ExG [11]	0,566	6,937	0,794	0,045
ExGr [11]	0,566	6,884	0,775	0,049
GLI [12]	0,457	12,563	0,624	0,297

Specifically, the optimized model achieved the lowest NRMSE of 0.457, indicating that the resulting NDVI image is close to the true value. The fine-tuned model also achieved the highest PSNR of 12.563 dB, indicating the highest quality and similarity of the NDVI image to the original image compared to the other methods. Fig. 11 illustrates the comparison of different vegetation indices from the satellite image.

In terms of feature similarity, the ORB value of the enhanced model reaches 0.794, comparable to the best synthetic method such as EXGR (0.775). In addition, the enhanced model also achieved the highest SSIM value of 0.297, indicating a high structural similarity between the generated NDVI image and the actual NDVI image. The SSIM value may vary which may occur when the generated NDVI image has a different contrast or inaccurate representation compared to the actual NDVI image [13].

The difference between SSIM and ORB values arises from their different operational focus. SSIM evaluates image quality based on perceived changes in structural information, taking into account luminance, contrast and structure, which is more in line with human visual perception. On the other hand, ORB is designed for fast keypoint detection and matching.

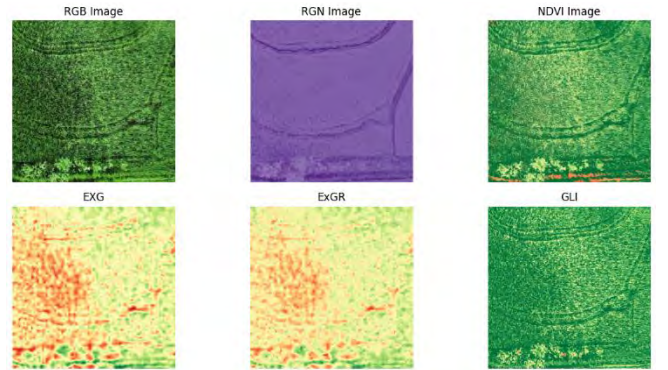


Fig. 11. Comparative visualization of vegetation indices from aerial imagery

F. Experimental Settings

In this research, agricultural land is used as a test object using the YOLOv5 method for object detection, which is then used to analyze plant health in RGB images using GLI. The tested farmland can be seen in Fig. 12.

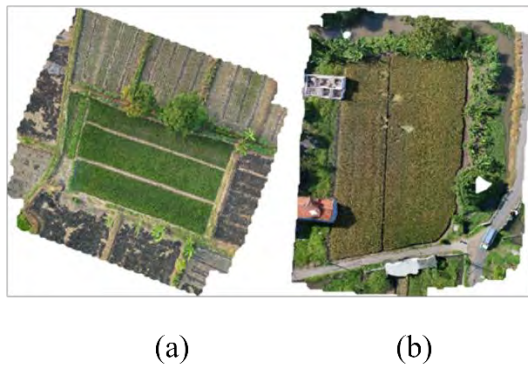


Fig. 12. Experimental settings (a) Gresik (b) Sidoarjo

YOLOv5 is used in this study to perform farmland segmentation and classification, with the aim of identifying and separating paddy fields from the rest of the background. After segmentation, a cropped image of the detected paddy field area can be retrieved for further analysis, such as crop health evaluation using methods such as GLI or synthetic NDVI analysis.



Fig. 13. Result YoloV5m (a) Gresik (b) Sidoarjo

GLI was used to generate a grayscale image highlighting the greenery area, while this synthetic NDVI was generated from the comparisons made as listed in Table 4. The GLI image was then converted into a binary image to separate healthy and unhealthy plant areas, with the unhealthy areas colored red in the original image for visualization purposes. The final analysis was interpreted to provide an overview of the plant health condition based on the percentage of area considered healthy.

TABLE V. RESULT PLANT HEALTHY CONDITION

Image	Plant Area (px)	Total Area (px)	Healthy Area (%)
(a) Gresik 0.92	21364043	26962584	79.24%
(a) Gresik 0.91	31590527	40743736	77.53%
(b) Sidoarjo 0.95	16787871	21030432	79.83%
(b) Sidoarjo 0.93	12067496	16213134	74.43%

V. CONCLUSION

This study found that complex YOLOv5 configuration models do not always provide the most accurate results. changes in epoch values, batch_size, yolo models can affect the model to be trained. More epochs can improve accuracy

but can also experience overfitting. In this research, the best model obtained is Yolo5m with epoch 50, batch 16, and the division of training data 70%, test data 20% and validation data 10%. The best height to detect agricultural land is 20 meters, although the results are not optimal. plant health can be measured using GLI which has similarities to NDVI. In testing the system, detecting agricultural land on agricultural land mapping obtained satisfactory results. We will also evaluate by integrating the model with drones as edge intelligence to find plant pests. Future work for this research will integrate of pest and disease classification, and time-series monitor monitoring for early detection.

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